PARAMETRIZING A POPULATION BALANCE EQUATION

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Abstract. Surely this cannot be that difficult.

1. Introduction

A plethora of quantities arise in this work, so we endeavour to consistently indicate their nature at a glance. All constants $\|\cdot\|, \langle\cdot\rangle, |\cdot| \in \overline{\mathbb{R}^+}$ are non-negative, and enclosed in $\|\cdot\|$ or $\langle\cdot\rangle$ without meaning (there are no norms or inner products here), or $|\cdot|$ only when taking the absolute value of $\cdot$. All *italic* quantities are dimensionless, to ease the unavoidable chore of tracking units. Undefined notation adheres to the Digital Library of Mathematical Functions [1].

A number $\mathrm{n}(\mathrm{t},\mathrm{L})$ (m^{-4}) of particles of length $\mathrm{L}\,(\mathrm{m})$ grow within each unit of reactor volume $\mathrm{V}\,(\mathrm{m}^3)$ as seconds $\mathrm{t}\,(\mathrm{s})$ pass. Conserving mass, the Population Balance Equation is

$$\begin{aligned} \partial_{\mathrm{t}}(\mathrm{Vn}(\mathrm{t},\mathrm{L})) &= -\mathrm{G}^+\partial_{\mathrm{L}}(\mathrm{Vn}(\mathrm{t},\mathrm{L})) \\ \mathrm{n}(0,\mathrm{L}) &= 0 \\ \mathrm{n}(\mathrm{t},0) &= \mathrm{B}^+/\,\mathrm{G}^+ \end{aligned} \tag{1.1}$$

where $\mathrm{B}^+\,(\mathrm{m}^{-3}\mathrm{s}^{-1})$ and $\mathrm{G}^+\,(\mathrm{m\,s}^{-1})$ are particle birth (nucleation) and growth rates. These relate empirically to supersaturation $S > 1$:

$$\begin{aligned} \text{birth} \quad \mathrm{B}^+ &:= \|\mathrm{B}\|\,(S-1)^{+|b|} \\ \text{growth} \quad \mathrm{G}^+ &:= \|\mathrm{G}\|\,(S-1)^{+|g|} \end{aligned} \tag{1.2}$$

where $(\#)^+ := \max(0,\#)$ denotes the positive part of $(\#)$. In experiments, critical supersaturation $S = 1$ might occur many times, but we cannot allow this. We must insist that

$$\text{supersaturation} \quad S \in (1, \sup S] \quad \text{at all times} \quad \mathrm{t} > \mathrm{t}^* \tag{1.3}$$

for finite supersaturation $\sup S \in \mathbb{R}$ and critical time $\mathrm{t}^* \in \mathbb{R}$ at which S first exceeds 1. This is not limiting per se, but it does mean that each critical instant in an experiment theoretically begins a new experiment. Throughout this work, an asterisk always denotes the critical instant $t = S - 1 = 0$

Given lengthscale $\|\mathrm{L}\|\,(\mathrm{m})$ and timescale $\|\mathrm{t}\|\,(\mathrm{s})$, dimensionless prefactors are defined as

$$\begin{aligned} \text{birth} \quad \|B\| &:= \|\mathrm{B}\|\,\|\mathrm{L}\|^3\,\|\mathrm{t}\| \\ \text{growth} \quad \|G\| &:= \|\mathrm{G}\|\,\|\mathrm{L}\|^{-1}\,\|\mathrm{t}\| \end{aligned} \tag{1.4}$$

together with dimensionless

$$\begin{aligned} \text{time} \quad t &:= \|\mathrm{t}\|^{-1}\,(\mathrm{t}-\mathrm{t}^*) \\ \text{volume} \quad V(t) &:= \|\mathrm{L}\|^{-3}\,\mathrm{V} \end{aligned} \tag{1.5}$$

This work aims to elucidate the dimensionless empirical birth and growth parameters $\|B\|, |b|, \|G\|, |g|$.

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2. The Crystal Size Distribution (CSD)

The Population Balance Equation conveys a symmetry between synthetic 'time' T and 'length' L defined as

$$S(t) := \max(S, 1) \tag{2.1a}$$

$$T := \|\mathrm{L}\|^{-1} \int_0^{\mathrm{t}} \mathrm{G}^+ \,\mathrm{dt} = \int_0^t \|G\| \left(S(\dot{t}) - 1\right)^{|g|} \mathrm{d}\dot{t} \tag{2.1b}$$

$$L := T - \|\mathrm{L}\|^{-1} \mathrm{L} \tag{2.1c}$$

These forms are chosen to render the PBE linear and invariant to exchanging T and L. This symmetry manifests in the particle count $N(T)f(T, L)$ defined as

$$N(\mathbb{R}^-) := \{0\} \tag{2.2a}$$

$$N(T) := \int_0^{\mathrm{t}} \mathrm{B}^+\mathrm{V} \,\mathrm{d\dot{t}} = \int_0^t \|B\| \left(S(\dot{t}) - 1\right)^{|b|} V(\dot{t}) \,\mathrm{d}\dot{t} \tag{2.2b}$$

$$f(T, L) := N(T)^{-1} \|\mathrm{L}\|^{-1} \mathrm{Vn(t, L)} \tag{2.2c}$$

Substituting these variables in Population Balance Equation (1.1) immediately yields

$$\begin{aligned} \partial_T N(T) f(T, L) &= 0 \\ N(T) f(T, L) &= \partial_t N(T) / \partial_t T = \partial_T N(T) \end{aligned} \tag{2.3}$$

These two lines combine to establish

$$\begin{aligned} f(T, L) &= N(T)^{-1} \partial_L N(L) \\ N(T)^{-1} N(L) &= \int_0^L f(T, \dot{L}) \,\mathrm{d}\dot{L} \end{aligned} \tag{2.4}$$

The Crystal Size Distribution (CSD) at 'time' T is $N(T)^{-1}N(L)$, its probability density $f(T, L)$, and its moments

$$\begin{aligned} M_0(T) &:= \int_0^T f(T, L)(T - L)^0 \,\mathrm{d}L \;\; = N(T)^{-1} N(T) = 1 \\ M_m(T) &:= \int_0^T f(T, L)(T - L)^m \,\mathrm{d}L = N(T)^{-1} m \int_0^T N(L) M_{m-1}(L) \,\mathrm{d}L \end{aligned} \tag{2.5}$$

The number of particles born by 'time' T is $N(T)$, and the number of particles whose 'length' does not exceed L is $N(L)$. This is the symmetry conveyed by the PBE: 'time' T and 'length' L are effectively the same thing. Like a growing child, a crystal equates how old it is with how big it is.

The CSD $N(T)^{-1}N(L)$ is the fraction of particles at 'time' T whose 'length' does not exceed L, which is the same as the fraction of particles at 'time' T born by 'time' L.

3. Taylor Approximation

This Section determines the CSD from supersaturation $S(t)$ and volume $V(t)$, expanded about $t = \bar{t}$ as

$$\begin{aligned} \ln(S(t) - 1) \;\; &= \sum_{k=0}^{K} \frac{\bar{\sigma}^{(k)}}{\Gamma(k+1)} (t - \bar{t})^k + O\left(\frac{\bar{\sigma}^{(K+1)}}{\Gamma(K+2)} (t - \bar{t})^{K+1} \right) \\ \text{where} \quad \bar{\sigma}^{(k)} &:= \partial_t^k \ln(S(\bar{t}) - 1) \\ V(t) \;\; &= \sum_{k=0}^{K} \frac{\bar{V}^{(k)}}{\Gamma(k+1)} (t - \bar{t})^k + O\left(\bar{V}^{(K+1)} \frac{(t - \bar{t})^{K+1}}{\Gamma(K+2)} \right) \\ \text{where} \quad \bar{V}^{(k)} &:= \partial_t^k V(\bar{t}) \end{aligned} \tag{3.1}$$

The partial derivatives here are normally provided by an exogenous model of supersaturation and mixing.

3.1 Linear Approximation

Taking $K = 1$ requires

$$\bar{\sigma}' > 0 \tag{3.2}$$

Integrating Equation (2.2b) using Equation (3.1) with $K = 1$ yields

$$\begin{aligned} \|N\|^{-1} N(T) &= \left(\bar{v} + \bar{V}'t\right) \exp\left(\|b\|\, t\right) - \bar{v} \\ \text{where} \quad \|b\| &:= |b|\, \bar{\sigma}' \\ \|\bar{\sigma}\| &:= |b|\, \bar{\sigma} - \|b\|\, \bar{t} \\ \|N\| &:= \|B\|\, \|b\|^{-1} \exp \|\bar{\sigma}\| \\ \bar{v} &:= \bar{V} - \bar{V}'\bar{t} - \bar{V}'\, \|b\|^{-1} \end{aligned} \tag{3.3}$$

Substituting Equation (3.1) with $K = 1$ in Equation (2.1) for small $|g|$

$$\partial_t L = \partial_t T = \|G\| + \frac{|g|}{|b|}\, \|G\|\, (\|b\|\, t + \|\bar{\sigma}\|) + O(|g|^2) \tag{3.4}$$

which integrates over $t \in [0, T]$ to yield

$$t = \frac{T}{\|G\|} - \frac{|g|}{|b|}\, \frac{T}{\|G\|} \left(\frac{\|b\|}{2}\, \frac{T}{\|G\|} + \|\bar{\sigma}\| \right) + O(|g|^2) \tag{3.5}$$

3.2 Quadratic Approximation

Taking $K = 2$, integration "completes the square", effectively shifting $\bar{t}$ to the maximum of $\ln(St - 1)$ so that

$$\bar{\sigma}' := 0 > \bar{\sigma}'' \tag{3.6}$$

Integrating Equation (2.2b) using Equation (3.1) with $K = 2$ yields

$$\begin{aligned} \|N\|^{-1} N(T) &= \bar{v}\, (\operatorname{erf}(\|b\|\, (t - \bar{t})) + \operatorname{erf}(\|b\|\, \bar{t})) - \left(\frac{\bar{V}''}{2} t + \bar{v}' \right) \exp\left(\|b\|^2\, t(2\bar{t} - t) \right) + \bar{v}' \\ \text{where} \quad \|b\| &:= \left| \frac{|b|\, \bar{\sigma}''}{2} \right|^{1/2} \\ \|\bar{\sigma}\| &:= |b|\, \bar{\sigma} - \|b\|^2\, \bar{t}^2 \\ \|N\| &:= \frac{\|B\| \exp \|\bar{\sigma}\|}{2\, \|b\|^2} \\ \bar{v} &:= \|b\|\, \pi^{1/2} \exp\left(\|b\|^2\, \bar{t}^2 \right) \left(\bar{V} + \frac{\bar{V}''}{4\, \|b\|^2} \right) \\ \bar{v}' &:= \bar{V}' - \frac{\bar{V}''\bar{t}}{2} \end{aligned} \tag{3.7}$$

Substituting Equation (3.1) with $K = 2$ in Equation (2.1b) for small $|g|$

$$\partial_t T = \|G\| + \frac{|g|}{|b|}\, \|G\| \left(\|b\|^2\, t(2\bar{t} - t) + \|\bar{\sigma}\| \right) + O(|g|^2) \tag{3.8}$$

which integrates to yield

$$t = \frac{T}{\|G\|} - \frac{|g|}{|b|}\, \frac{T}{\|G\|} \left(\|b\|^2\, \frac{T}{\|G\|} \left(\bar{t} - \frac{1}{3}\, \frac{T}{\|G\|} \right) + \|\bar{\sigma}\| \right) + O(|g|^2) \tag{3.9}$$

References

1. *NIST Digital Library of Mathematical Functions*, `https://dlmf.nist.gov/`, Release 1.2.6 of 2026-03-15, F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.